

How Much of Truth Fits on a Single Axis?

Knowledge-Dependent Dimensionality, Unsupervised Polarity Recovery,
and Seven Falsification Attempts in a Large Language Model

Francesco Karim Vicidomini

Independent researcher

Reference code: `truth_probe.py`, `canvas.py` — Model: Qwen2.5-1.5B

June 28, 2026

Abstract

Two dominant families of methods read factual truth from the hidden states of language models. *Linear probes* project the residual stream onto a single learned or difference-of-means axis; *sparse autoencoders* (SAEs) expand the state against an overcomplete dictionary to disambiguate features in superposition. The first is cheap but treats truth as a static one-dimensional axis; the second is interpretable but costly. We study a third, deliberately minimal object: an *unsupervised* geometric probe obtained from the singular value decomposition (SVD) of paired true/false differences, readable at cost $O(d)$ per token with no training. On Qwen2.5-1.5B, under a held-out cross-validation protocol with a label-permutation null, we report three findings. (i) The unsupervised axis carries a *real* truth signal—held-out AUC up to 0.938 on clean minimal pairs, permutation $p = 0.005$ —but a full linear probe on the same raw states reaches 0.989; the gap is informative rather than incidental. (ii) That gap *scales with how well the model knows the fact*: ≈ 0.05 on facts the model plausibly knows, ≈ 0.20 on facts it plausibly does not, which we read as a near one-dimensional truth representation when the fact is known and a diffuse one when it is not. (iii) On mixed-polarity data the unsupervised SVD *recovers* the supervised polarity direction of Bürger et al. at cosine 0.959, without ever seeing the polarity label. We then ask whether the one-dimensional reading is real or merely the easiest thing to see, and subject it to *seven independent falsification attempts*—a complex/phase second dimension, a global depth signature, a dynamic “butterfly effect” propagation, a depth-department decomposition, an energy coordinate, a phase coordinate, and a three-dimensional class-clustering—each run with the same shield. None overturns the claim: the depth signature is statistically real but never practically beats a single layer (+0.015 to +0.019 AUC); the energy and phase coordinates separate at chance; and in three dimensions true and false do not form distinct clumps (centroid ratio 0.48–0.55). What survives is precise: truth in this model is one-dimensional, carried by the position axis, concentrated by depth, and its separability scales with how well the model knows the fact (CounterFact 0.795 vs. TruthfulQA 0.656). We read a result that resists seven independent falsifications as strengthened, not weakened, and interpret the overlap on unknown facts as the observable signature of unresolved superposition. Results are preliminary (single model, small datasets) and intended to be falsifiable.

1 Introduction

Large language models produce false statements with the same fluency as true ones. Predicting when a model is about to leave the truth—ideally during generation, without a second judge model—is a central reliability problem. Mechanistic interpretability has shown that many high-level properties, including the truth value of a factual statement, are encoded as linear directions in the residual

stream [2, 4]. Two method families follow. The first is the *linear probe*: a classifier or difference-of-means axis whose projection separates true from false [1, 2, 3]. The second is *sparse dictionary learning*: project the state onto an overcomplete dictionary, potentially millions of atoms, to undo superposition and read monosemantic features [6, 7].

Each pays a price. A single static axis is fragile and low-capacity; a sparse autoencoder requires dedicated training and a large matrix multiply per token. This paper takes a third stance. Rather than learning a probe or expanding a dictionary, we *read structure from geometry alone*, and ask a precise question: *how much of the truth structure does a single direction capture, and when does that capture fail?* We extract all hidden states in one forward pass, let an axis emerge without supervision from the SVD of paired true/false differences, and then—crucially—measure everything out of sample, with a permutation null that quantifies how much apparent accuracy is mere selection. The answer is modest but, we believe, clean: the axis reads truth cheaply when the fact is known, fails predictably when it is not, and—blind to the label—recovers the polarity structure that a prior supervised method constructed by hand.

A single positive result invites the obvious objection: is the one-dimensionality real, or merely the easiest thing to see? The second half of this paper answers by attack. We subject the one-dimensional claim to seven independent falsification attempts, each designed to expose a richer structure if one exists, and report that none overturns it. We regard a result that survives seven such attempts as strengthened.

Contributions.

1. An unsupervised, training-free truth probe at cost $O(d)$, with a rigorous held-out and permutation-tested evaluation that corrects the optimism of in-sample axis selection (Sec. 4.1).
2. Evidence that the *dimensionality of the truth representation is knowledge-dependent*: near one-dimensional for known facts, diffuse for unknown ones, quantified by the gap to a full linear probe (Sec. 4.2).
3. *Unsupervised recovery of the polarity direction* t_P of Bürger et al. [5]: the SVD of mixed-polarity differences aligns with the supervised t_P at cosine 0.959, with no polarity label (Sec. 4.4).
4. A *robustness battery*: seven independent falsification attempts against the one-dimensional reading—second dimension, depth-global structure, dynamics, departments, energy, phase, three-dimensional clustering—each tested under held-out CV and a permutation null, none of which overturns the claim (Sec. 6).
5. An interpretation of the residual overlap on unknown facts as the observable signature of unresolved superposition, tied to a capacity argument (Secs. 5, 8).

2 Related work

Linear truth directions. [1] extract a knowledge direction without labels by enforcing logical consistency. [2] show truth has emergent linear structure and that difference-of-means probes generalize as well as more complex ones. [3] intervene along truthful directions at inference time; [4] frame representation reading top-down. Our axis shares the linear-direction hypothesis but is read from the geometry of paired differences rather than imposed by a trained probe.

A two-dimensional truth subspace. Closest to us, Bürger et al. [5] show that truth is represented not in a single direction but in a two-dimensional subspace spanned by a general truth direction t_G and a polarity-sensitive direction $t_P \sim \text{XOR}(\text{is_true}, \text{is_negated})$, and that probes trained on affirmative statements fail on negated ones. We confirm this structure in Qwen2.5-1.5B and show, as a methodological addition, that t_P can be recovered *without* polarity supervision.

Cross-layer dynamics. Recent monitors model the evolution of hidden states across depth rather than a single layer [9, 10]. Our depth trajectories are descriptive and consistent with this line.

Superposition and dictionaries. Superposition [8] motivates decomposing the residual stream into overcomplete dictionaries [6, 7]. This is the operation we explicitly do *not* perform; Secs. 5 and 8 argue why a single projection cannot substitute for it.

3 Method

3.1 Extraction and the unsupervised axis

We load Qwen2.5-1.5B frozen (28 blocks, hence 29 hidden-state levels: the embedding plus each block output, $d = 1536$) and, per sentence, run one forward pass with `output_hidden_states`, taking the last-token state at every layer. For N sentences this yields $H \in \mathbb{R}^{N \times 29 \times d}$.

The data are *minimal pairs*: each true sentence is matched to its false counterpart with the same subject and a flipped predicate (e.g. “The Sun is a star.” vs. “The Sun is a planet.”). Within a pair, the difference cancels the shared topic and leaves the truth signal. We form the intra-pair difference matrix D with rows $h_{\text{true}} - h_{\text{false}}$, and take its SVD,

$$D = USV^\top, \quad v_1 = V_1, \tag{1}$$

where v_1 is the dominant direction of the differences. The labels do not build v_1 ; the pairing does. A single orientation bit (“true is positive”), fit on training data, fixes the sign.

3.2 A falsifiable evaluation protocol

The central methodological point is that the axis, its calibration, the orientation bit, *and* the choice of peak layer must be fit on training data only. We therefore use k -fold cross-validation *over pairs* (a pair is never split across folds, which would leak the shared topic), fit everything on the training folds, and report held-out AUC averaged across folds. To quantify how much apparent accuracy is selection over the 29 layers, we run a *permutation null*: we randomly swap true/false *within* each pair (labels become noise, the pair structure is preserved) and recompute the maximum held-out AUC over layers; the p -value is the fraction of the null at or above the observed value. As a reference point we also fit an ℓ_2 -regularized logistic probe on the raw d -dimensional states, held out over the same pair folds.

3.3 The complex-plane extension we test

We additionally form a second axis v_2 , the SVD residual orthogonal to v_1 , treat the projection as $z = c_1 + i c_2$ with calibrated real and imaginary parts, and test whether magnitude- or phase-based scores improve separation over the real axis alone. They do not (Sec. 4.3); we keep the result because the negative finding is itself informative.

Table 1: Held-out AUC (5-fold CV over pairs) at layer 16, Qwen2.5-1.5B. “1D” is the real axis; “MAG” the magnitude-gated polar score; “PHASE” the angular-deviation score; “2D” a logistic on (Re, Im); “Probe” an ℓ_2 logistic on raw states. The last column is the label-permutation p -value for the 1D axis.

Dataset	pairs	1D	MAG	PHASE	2D	Probe	p
Clean minimal pairs	45	0.938	0.936	0.928	0.931	0.989	0.005
CounterFact / TruthfulQA mix	250	0.638	0.640	0.676	0.651	0.841	0.005

3.4 Polarity design

To test whether a genuine second dimension exists, we use a 2×2 design: each fact in four forms—affirmative-true (AT), affirmative-false (AF), negated-true (NT), negated-false (NF)—giving the XOR structure of [5]. We compute the affirmative and negated truth directions $t_A = \overline{AT} - \overline{AF}$, $t_N = \overline{NT} - \overline{NF}$, the general direction $t_G \propto t_A + t_N$, and the supervised polarity direction $t_P \propto t_A - t_N$. For the unsupervised recovery test, we run the difference SVD on minimal pairs that *mix* the two polarities, so the SVD never sees the polarity label, and compare its leading directions to the supervised t_P .

4 Results: signal, dimensionality, polarity

All numbers are on Qwen2.5-1.5B with last-token pooling; AUC is out-of-fold unless stated.

4.1 A real but bounded truth signal

Table 1 reports the held-out evaluation at the peak layer (layer 16). On clean minimal pairs the unsupervised axis reaches AUC 0.938 with permutation $p = 0.005$ (null mean 0.588, 95th percentile 0.664): the geometric axis aligns with truth well beyond what layer selection alone can manufacture. The signal is therefore real. It is also *bounded*: a full linear probe on the same raw states reaches 0.989. The probe sees information the single axis discards.

4.2 The gap scales with model knowledge

The two rows of Table 1 differ not only in difficulty but in the *size of the probe gap*. On clean facts the model plausibly knows, the unsupervised axis (0.938) trails the full probe (0.989) by only ≈ 0.05 . On CounterFact—which contains many obscure relations a 1.5B model plausibly does not know (e.g. the mother tongue of a minor historical figure)—the axis (0.638) trails the probe (0.841) by ≈ 0.20 . The signal that the single axis cannot recover is precisely the part that lives off-axis, distributed across many directions.

We read this as a statement about *dimensionality*. When the model knows a fact, its truth content is concentrated and a single direction captures most of it. When the model does not know a fact, the truth-relevant signal is weak and diffuse, and one direction recovers only a fraction. The dimensionality of the truth representation is thus not fixed but *knowledge-dependent*. This is, to our knowledge, a framing that neither the linear-probe nor the dictionary-learning literature states in these terms, and it is the contribution we consider most novel.

Table 2: Polarity analysis (leave-facts-out CV), layer 14 chosen on affirmative data; and unsupervised t_P recovery at layer 15. AUC below 0.5 indicates a sign flip.

Quantity	Value
$\cos(t_A, t_N)$	-0.415
Affirmative axis on affirmative (held-out AUC)	0.981
Affirmative axis on <i>negated</i> (held-out AUC)	0.079
General direction t_G on negated (held-out AUC)	0.954
Unsupervised SVD vs. supervised t_P : $\max_k \cos(\text{SVD}_k, t_P) $	0.959
(principal component carrying t_P)	PC 1
$\cos(\text{SVD}_1, t_G)$	0.157

4.3 The complex plane does not help; “phase” is the sign flip

On single-polarity data the second dimension adds nothing: at layer 16 the 2D logistic (0.931) does not exceed the 1D axis (0.938). This is structural. Minimal pairs cancel the topic, so the differences concentrate along one direction; the SVD places the entire signal in v_1 , and v_2 is residual variance—noise, not a second truth factor. A controlled synthetic check confirms that the apparent “rotation” of falsehoods is an artifact: with a pure sign flip and zero genuine rotation, an angular (phase) score still separates classes, because a state at $\text{Re} < 0$, $\text{Im} \approx 0$ has phase $\approx \pi$ by definition. The earlier intuition of truth-as-phase, in the manner of a critical line, has no operational content here: it reduces to the sign of the real axis re-expressed in polar coordinates. We therefore retract that framing.

4.4 Unsupervised recovery of the polarity direction

A genuine second dimension does exist, but only when the data span both polarities, and it is the polarity direction of [5], not a phase. Table 2 shows the diagnostic: an axis fit on affirmative statements classifies affirmative held-out facts at 0.981 but *anti-classifies* negated ones at 0.079—a near-total flip—while the general direction t_G recovers the negated facts at 0.954. The truth direction genuinely rotates with polarity ($\cos(t_A, t_N) = -0.415$).

The methodological result is the recovery. When we run the unsupervised difference SVD on *mixed-polarity* minimal pairs—blind to the negation label—its leading direction aligns with the supervised polarity direction t_P at cosine 0.959 (layer 15), and t_P lands in the first principal component, while alignment with t_G is only 0.157. In other words, when both polarities are present, the dominant axis of the differences *is* polarity, and the SVD finds it without supervision. Where [5] constructs t_P by separating affirmative from negated statements by hand, the same direction emerges here from geometry alone.

5 Discussion: what a single axis can and cannot read

It is tempting to view a cheap geometric probe as a low-cost version of dictionary learning—reading superposition without expanding it. We argue this is impossible, and that the impossibility is a capacity statement, not an implementation gap. A projection is an inner product: it collapses a d -dimensional state to a single scalar, discarding $d - 1$ dimensions. From that scalar one cannot reconstruct which features were present, only how much a chosen direction was. “Reading the superposition” requires asking the question once per dictionary atom, which over an overcomplete

dictionary of size m is exactly the $O(dm)$ sparse autoencoder. Compression (project to a scalar) and disambiguation (expand to a sparse code) are not inverse cheap-vs-expensive versions of one operation; they are different operations, and only the second reads superposition. The ≈ 0.20 probe gap of Sec. 4.2 is a direct measurement of the truth information that a single axis structurally cannot return.

The consequence is a clear scope. This method’s value is *structural and cheap*: a training-free axis that, on facts the model knows, rivals a trained probe at two dot products per token; a quantitative law relating the readability of truth to the model’s knowledge of the fact; and an unsupervised route to the polarity direction. It is not a substitute for SAEs and should not be presented as one.

6 Robustness: seven falsification attempts

The first half of this paper established a positive claim. We now stress-test it. The claim under test is:

In Qwen2.5-1.5B, the truth value of a factual statement is carried by a single (one-dimensional) direction—the position axis obtained without supervision from the SVD of paired true/false differences—on facts the model knows; the second and higher dimensions add no separation.

A claim of one-dimensionality is falsifiable in a specific way: exhibit a second, third, or depth-distributed coordinate that separates true from false better than the single axis. We construct seven such coordinates and test each.

Two kinds of test, kept distinct. We separate *controlled experiments* from *visual observations*, and label every result as one or the other. Controlled experiments (attempts 1–4) produce a held-out AUC under k -fold cross-validation *over pairs* (a pair is never split across folds) and a label-permutation null (true/false swapped within pairs) that quantifies selection bias; a result is meaningful only if it clears that null. Visual observations (attempts 5–7) do not classify; they project the states and measure a single descriptive statistic—the per-coordinate separation AUC, or the centroid-separation ratio—so the conclusion follows from a number attached to a picture, not from a fitted model. Throughout, the model is frozen Qwen2.5-1.5B, last-token pooling, peak layer 16 unless stated.

Table 3 summarizes all seven with their real numbers; the prose below gives the reasoning for each.

1 – The complex plane. The original framing promoted the second SVD direction to an imaginary axis and read truth as a phase rotation. If real, a phase- or magnitude-based score should beat the real axis. It does not: on clean minimal pairs the phase score reaches AUC 0.928 against 0.938 for the real axis, and a controlled synthetic check shows that with a pure sign flip and *zero* genuine rotation an angular score still separates, because a state at $\text{Re} < 0$ has phase $\approx \pi$ by definition. The “rotation” is the sign of the real axis re-expressed in polar coordinates. The complex framing is decorative; we retract it. (This is the same negative result reported in Sec. 4.3, restated here as the first falsification attempt.)

2 – A global depth signature. If truth flows through the network, the full trajectory $[\Delta_0, \dots, \Delta_{28}]$ of the signed deviation on a fixed axis should beat the single best layer. On clean pairs it does

Table 3: Seven attempts to falsify the one-dimensional truth axis. “Exp.” = controlled experiment (held-out CV + permutation null); “Obs.” = visual observation (descriptive separation statistic). None overturns the claim.

#	Richer structure hypothesized	Type	Result (real numbers)	Verdict
1	A complex/phase second dimension (truth as rotation)	Exp.	PHASE AUC 0.928 vs. 1D 0.938 (clean); reduces to the real-axis sign flip	falsified
2	Truth is global across depth (fixed-axis signature)	Exp.	global 0.923 vs. local 0.938 (−0.015), $p=0.0099$; persistence gap +0.04	falsified
3	Truth is dynamic—a “butterfly effect” in the unsaturated shade, on hard data	Exp.	global 0.642 vs. local 0.627 (+0.015), $p=0.0099$; commitment true 2.83→5.91 vs. false 2.65→5.86	falsified
4	Truth is distributed across layer “departments”	Exp.	diagnostic curve is a smooth ramp (no steps); global 0.646 vs. local 0.627 (+0.019), $p=0.0099$	falsified
5	The argument θ (phase) carries truth	Obs.	θ AUC 0.504 / 0.584 / 0.556 (CF / TQA / mix)	falsified
6	The energy $ z $ (commitment) carries truth	Obs.	$ z $ AUC 0.560 / 0.558 / 0.543	falsified
7	True/false form distinct 3D clumps	Obs.	centroid ratio 0.55 / 0.51 / 0.48 (< 1, overlapping); Re alone 0.795 / 0.656	falsified

not: the depth-signature classifier reaches held-out AUC 0.923 against 0.938 for the single layer (−0.015), and although the signature is real (permutation $p = 0.0099$), it carries no advantage. The crossing-persistence gap between false and true is +0.04—flat. No propagation.

3 – A dynamic butterfly effect in the shade. The signed deviation saturates (σ compresses large values to $\pm\frac{1}{2}$), so a propagating *uncertainty* could be invisible to it. We re-ran on hard data (CounterFact/TruthfulQA mix) with the unsaturated features [Re, $|z|$]*—the “shade.”* The depth signature now beats the single layer, but by +0.015 (0.642 vs. 0.627, $p = 0.0099$): real and marginal. The decisive descriptive measure kills the butterfly effect: the commitment magnitude grows identically for both classes (true 2.83 → 5.91, false 2.65 → 5.86); false statements do not hesitate more than true ones, and there is no propagating divergence.

4 – Layer departments. If different layer bands perform different jobs (surface, semantics, logic), a single semantic axis re-read everywhere wastes most layers; giving each layer its *own* native probe should recover a distributed signal, and the per-layer separation curve should show steps. It shows a smooth ramp instead—the per-layer native separation rises continuously to a single peak at layer 16, with no plateaus and no jumps. The per-layer signature beats the single layer by +0.019 (0.646 vs. 0.627, $p = 0.0099$)—again real and marginal. No departments; the native-axis “cure” improves the fixed axis but does not change the ceiling.

5–6 – Energy and argument. Decomposing each state into position (Re), energy ($|z|$), and argument (θ) and measuring how well each separates true from false (Fig. 1): only position carries

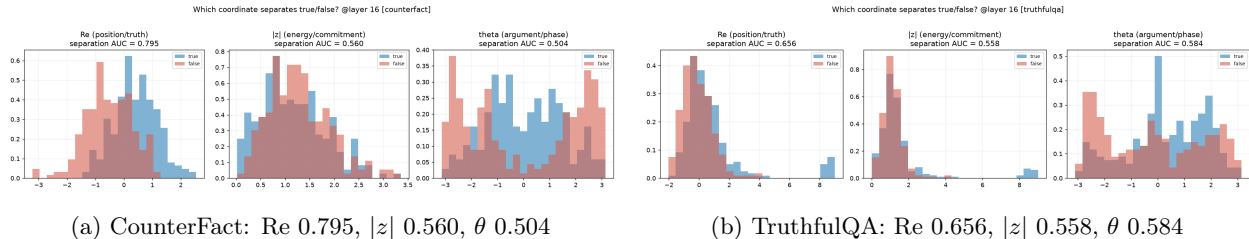


Figure 1: Attempts 5–6 and knowledge scaling, layer 16. Per-coordinate separation AUC. On both datasets only the position axis (Re) separates true from false; energy ($|z|$) and argument (θ) are at chance. The position separation falls from 0.795 (CounterFact, often-known facts) to 0.656 (TruthfulQA, more ambiguous), while the one-dimensional structure is unchanged.

signal. The energy separates at $|z|$ AUC 0.560/0.558/0.543 (CounterFact/TruthfulQA/mix)—true and false are equally committed. The argument separates at θ AUC 0.504/0.584/0.556—essentially chance; what little it shows is the echo of the Re sign, not independent information. The phase, the very coordinate the complex framing made central, is empty.

7 – Three-dimensional clumping. If truth were a richer object, true and false should form distinct clusters in the three-dimensional (v_1, v_2, v_3) space. They do not (Fig. 2). The centroid-separation ratio—distance between the true and false centroids divided by the average cloud spread—is 0.55/0.51/0.48, all below 1: the clouds overlap. Crucially this low ratio is *not* evidence against the axis; it is the one-dimensional signal seen through two noise dimensions. Projected onto position alone, the same data separate at AUC 0.795 (CounterFact). Adding the orthogonal dimensions dilutes, it does not enrich.

6.1 What survives

Three statements resist all seven attempts.

Truth is one-dimensional and lives on the position axis. Across every decomposition, only Re separates; energy, phase, the second and third SVD directions, and the full-depth trajectory add at most +0.02 AUC and usually nothing.

Separability scales with knowledge. On CounterFact, where the model often knows the fact, the position axis separates at AUC 0.795; on the more ambiguous TruthfulQA at 0.656; on the diffuse mix the centroids nearly coincide. The clearer the model’s knowledge, the more one-dimensional and separable the truth representation.

On unknown facts, the classes show low separability. Where the model does not know the fact, true and false representations do not form distinct clusters but overlap in space—intuitively a dense, uniform distribution rather than two distinct clumps. We state this as *low class separability / distribution overlap* (centroid ratio < 1), the measured counterpart of the knowledge-dependent dimensionality of Sec. 4.2.

7 The axis reads truth and confidence

The seven attempts establish that there is no second coordinate of truth to find. This has a constructive consequence that is easy to miss, so we state it plainly. The same unsupervised axis admits *two* readings, not one.

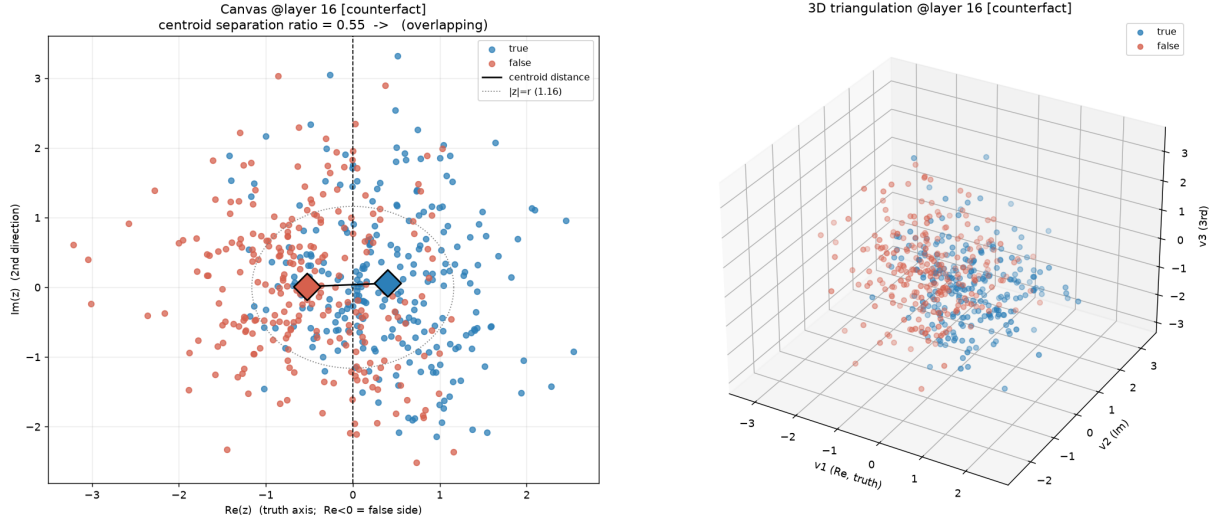


Figure 2: Attempt 7, CounterFact at layer 16. Left: the 2D plane; the true (blue) and false (red) centroids (diamonds) are close relative to the cloud spread (centroid-separation ratio 0.55, overlapping). Right: the three-dimensional (v_1, v_2, v_3) scatter shows a gradient along v_1 (position) but no distinct clumps. The separation is confined to the horizontal position axis; the extra dimensions are noise.

Its *sign* classifies a statement: $\text{Re} > 0$ is the true side, $\text{Re} < 0$ the false side. Its *separability*—how cleanly true and false split on a given set of facts—measures how well the model knows that material. The geometry that answers “is this statement true?” therefore also answers, indirectly, “does the model know this?”, because a model that does not know produces overlapping rather than separated representations (Figs. 2, 3). These are not two axes; they are two readings of one axis—the direction of the arrow, and the sharpness of the arrow.

The reading is quantitative at the level of a distribution. The position-axis separation is 0.795 on CounterFact and 0.656 on TruthfulQA: by this measure the model “knows” the CounterFact material better than the TruthfulQA material, read off the geometry alone, with no external grading. We report this as a property of the dataset, not yet of the single fact. Calibrating a per-fact confidence—a local separability around one statement that predicts whether the model actually errs on it—is the natural next step, and requires a behavioral label (does the model answer the open question correctly?) that the minimal-pair design does not contain. We mark it as future work rather than a claim, to keep the boundary between what is measured and what is conjectured explicit. Section 9 returns to this label and to a prerequisite it exposed.

8 Why unknown facts overlap: the signature of superposition

It remains to interpret *why* the classes overlap on unknown facts rather than simply asking *that* they do. The overlap is not measurement noise; it is the observable signature of unresolved superposition, and the explanation follows from the capacity argument of Sec. 5.

A projection onto a single direction returns a scalar: it collapses the d -dimensional state to one number and discards $d - 1$ dimensions. When the model knows a fact, its truth content has been allocated a clean, dominant direction of its own—the residual stream has, in effect, dedicated an axis to it—and the single projection recovers almost all of it (position AUC 0.795, near the full-probe ceiling). When the model does not know a fact, there is no dedicated direction: the weak

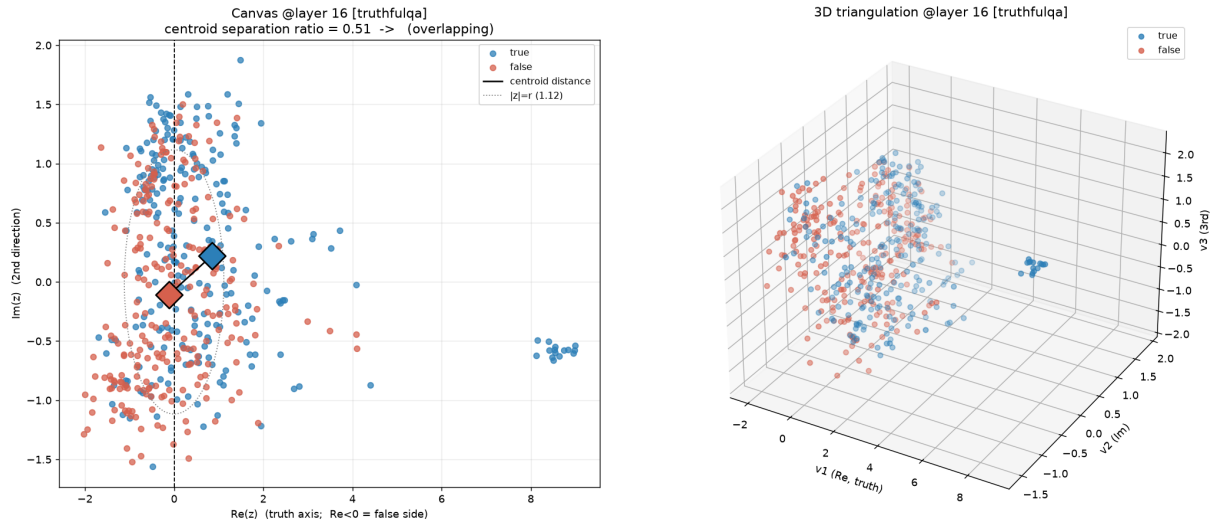


Figure 3: Attempt 7, TruthfulQA at layer 16. The same picture on more ambiguous facts: centroid-separation ratio 0.51 (overlapping), no three-dimensional clumps. Lower knowledge, lower separability, identical one-dimensional structure.

truth-relevant signal remains distributed across many superposed features, and no single axis can pull it out of the tangle. Recovering it would require asking the question once per dictionary atom—the expansion that a sparse autoencoder performs at cost $O(dm)$ —which is a different operation from a single projection, not a cheaper version of it.

The consequence is exactly what the canvas shows. On known facts the truth occupies its own direction, the classes form two regions, and the centroid ratio is high. On unknown facts the truth is spread across the superposition, the classes overlap, and the centroid ratio falls below one. The dense, uniform overlap of true and false on unknown facts is therefore not a failure of the probe but a direct, visual readout of superposition that the probe structurally cannot resolve. The ≈ 0.20 probe gap of Sec. 4.2 and the low centroid ratio measured here are two views of the same quantity: the truth information that lives off-axis, in superposition, on facts the model does not know.

9 A note for future work: verify behavior before geometry

Two natural extensions—a per-fact behavioral confidence label (Sec. 7) and a study of the geometry of *logical* (as opposed to semantic) truth—share a prerequisite that is easy to overlook, and we record it here as methodological guidance rather than as a result.

An informal behavioral check (not a controlled experiment). Before building any geometric study of a capability, one should confirm that the capability exists behaviorally. We ran a small, qualitative probe of the base model in open-ended completion (greedy decoding, no context carried between prompts). Two things were apparent on a handful of prompts. First, the model reproduces the common semantic facts used throughout this paper—capitals, chemical symbols, authorship, currencies—consistent with the “facts the model knows” assumption. Second, on symbolic transitive syllogisms built from nonce words (“a *grok* is heavier than a *blarn*; a *blarn* is heavier than a *fendu*; therefore ...”), the base 1.5B model does *not* perform the deduction: it deflects into multiple-choice or code (e.g. `SymPy`) scaffolding rather than completing the transitive conclusion. We stress that this

is an informal observation on a few prompts, not a measured result, and we draw no quantitative claim from it.

The principle. Studying the geometry of a capability presupposes the capability. Measuring the hidden-state geometry of a task the model cannot perform yields noise, and that noise is easily mistaken for “the property is distributed/complex” when the truthful reading is “the property is absent.” A cheap behavioral feasibility check should therefore precede any geometric study of *logical* truth; on this model that study is not feasible, which is itself useful to know in advance. The same principle applies to transfer: before studying the geometry of truth in a second model, one should first verify behaviorally that the second model knows the facts in question.

A hypothesis to test, not a claim. The way the base model deflects on uncertain input—falling back on memorized *format* scaffolding (exam layouts, question/answer templates, code)—suggests, but does not establish, that part of the overlap on unknown facts may carry a sentence-format component rather than pure superposition noise. This is consistent with the format confound that makes mixing datasets inadvisable (Sec. 11). We flag it as a direction for a future, properly shielded experiment, not as evidence.

10 Reproducibility

Two scripts reproduce every number in this paper. `truth_probe.py` runs the controlled experiments (the positive results and attempts 1–4); `canvas.py` renders the visual observations (attempts 5–7). Each result in `truth_probe.py` is an *isolated* subcommand: running one task loads, extracts, and computes only its own data, and touches nothing belonging to the others, so a reviewer who wants to verify one figure runs exactly one command. Both scripts load Qwen2.5-1.5B with safetensors only and load external datasets as Parquet only, never with remote code. A CUDA GPU is optional; on CPU the script runs in float32.

Both scripts are archived on Zenodo (DOI 10.5281/zenodo.20938285) and mirrored at github.com/FrancescoMarhel/TruthProbe.

Setup.

```
pip install torch transformers datasets
```

Positive results (Tables 1, 2).

```
# Table 1, top row (clean minimal pairs): held-out AUC, ablation,
#                               full-probe baseline, permutation p-value
python truth_probe.py signal --dataset builtin --with-2d --baseline --perm 200
```

```
# Table 1, bottom row (CounterFact / TruthfulQA mix)
python truth_probe.py signal --dataset mix --max-pairs 250 \
    --with-2d --baseline --perm 200
```

```
# Table 2, top (affirmative -> negated flip; the 2D polarity structure)
python truth_probe.py polarity
```

```
# Table 2, bottom (unsupervised SVD recovers the polarity direction t_P)
```

```
python truth_probe.py recovery
```

Controlled falsification attempts (Table 3, 1–4).

```
# attempt 1 (PHASE ablation) is the PHASE column of ‘signal’ above
```

```
# attempt 2: global depth signature (fixed axis), clean pairs
```

```
python truth_probe.py domino --dataset builtin --perm 100
```

```
# attempt 3: dynamic shade on hard data
```

```
python truth_probe.py domino --dataset mix --max-pairs 250 --signal-mag --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
    --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

```
# attempt 4: per-layer native axes + diagnostic ramp
```

```
python truth_probe.py domino --dataset mix --max-pairs 250 --per-layer-axis --perm 100 \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e \
    --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Visual observations (Figs. 1–3, attempts 5–7).

```
python canvas.py --dataset counterfact --layer 16 --out counterfact \
    --rev-counterfact c945b082ca08d0a8f3ba227fb78404a09614c36e
```

```
python canvas.py --dataset truthfulqa --layer 16 --out truthfulqa \
    --rev-truthfulqa 741b8276f2d1982aa3d5b832d3ee81ed3b896490
```

Each `canvas.py` run writes `<out>_coords.png` (per-coordinate separation), `<out>_plane.png` (the 2D plane with centroids and the separation ratio), and `<out>_3d.png` (the three-dimensional scatter). Upload the six PNGs alongside the `.tex` so the figures render. Run CounterFact and TruthfulQA *separately*; do not pass `-dataset mix` to `canvas.py`, as mixing injects a spurious sentence-format axis.

Data provenance and safety. The clean minimal pairs and the 2×2 polarity set are embedded in `truth_probe.py`. The external datasets are NeelNanda/`counterfact-tracing` (commit `c945b08...`) and `truthful_qa`, *generation* config (commit `741b827...`), loaded as Parquet only. For bit-exact reproduction the dataset commits may be pinned as shown, or a local file passed with `-file-counterfact <path>`. Add `-h` to any subcommand for its options.

11 Limitations

These results are preliminary and confined to a single, small model (Qwen2.5-1.5B); the entire battery—the positive results and all seven falsification attempts—should be re-run on a larger model and a different family before the one-dimensionality is treated as general. It is plausible that departments, dynamics, or a genuine second dimension emerge with scale, precisely because a 1.5B residual stream may be too compact to separate them. The datasets are small (tens to a few hundred pairs), so confidence intervals are wide despite the permutation control. The “unknown-fact” interpretation of the CounterFact gap and of the low separability is plausible but confounded: a fact the model does not know and a fact whose truth is genuinely multidimensional are not separated

here, and the low-separability reading is further confounded with sentence-format effects—which is why mixing datasets must be avoided, as it injects a spurious format axis. Disentangling ignorance from genuine multidimensionality (e.g. by filtering CounterFact to facts the model can answer, using the behavioral label of Sec. 9) is necessary to make the dimensionality claim quantitative. We use last-token pooling; other aggregations may shift the axis.

12 Conclusion

Stripped of a decorative complex-plane framing, the positive results survive on a real model under a falsifiable protocol: an unsupervised truth axis carries a permutation-tested signal at cost $O(d)$; the gap between that axis and a full probe scales with how well the model knows the fact, suggesting a knowledge-dependent dimensionality of truth; and the unsupervised difference SVD recovers the supervised polarity direction of [5] at cosine 0.959. The one-dimensional reading then withstands seven independent falsification attempts—spanning a second dimension, depth-global structure, dynamics, departments, energy, phase, and three-dimensional clustering—none of which exhibits a richer separating coordinate. We read such a result as strengthened, not weakened, by the absence of richer structure, and we interpret the residual overlap on unknown facts as the visual signature of unresolved superposition that a single projection structurally cannot resolve. The aim is a light, readable, almost-free structural probe to sit alongside generation—honest about the boundary it cannot cross. Natural next steps are a multi-model study of the dimensionality law, a knowledge-controlled dataset built from a behavioral label, and a benchmarked evaluation of the polar-gate monitor, each preceded by a behavioral feasibility check of the kind described in Sec. 9.

References

- [1] C. Burns, H. Ye, D. Klein, J. Steinhardt. *Discovering Latent Knowledge in Language Models Without Supervision*. arXiv:2212.03827, 2022 (ICLR 2023).
- [2] S. Marks, M. Tegmark. *The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets*. arXiv:2310.06824, 2023 (COLM 2024).
- [3] K. Li, O. Patel, F. Viégas, H. Pfister, M. Wattenberg. *Inference-Time Intervention: Eliciting Truthful Answers from a Language Model*. NeurIPS, 2023.
- [4] A. Zou et al. *Representation Engineering: A Top-Down Approach to AI Transparency*. arXiv:2310.01405, 2023.
- [5] L. Bürger, F. A. Hamprecht, B. Nadler. *Truth is Universal: Robust Detection of Lies in LLMs*. arXiv:2407.12831, 2024 (NeurIPS 2024).
- [6] H. Cunningham, A. Ewart, L. Riggs, R. Huben, L. Sharkey. *Sparse Autoencoders Find Highly Interpretable Features in Language Models*. arXiv:2309.08600, 2023.
- [7] T. Bricken et al. *Towards Monosemanticity: Decomposing Language Models With Dictionary Learning*. Transformer Circuits Thread, Anthropic, 2023.
- [8] N. Elhage et al. *Toy Models of Superposition*. Transformer Circuits Thread, Anthropic, 2022 (arXiv:2209.10652).

- [9] *ICR Probe: Tracking Hidden State Dynamics for Reliable Hallucination Detection in LLMs*. arXiv:2507.16488, 2025.
- [10] *LLM Hallucination Detection: A Fast Fourier Transform Method Based on Hidden Layer Temporal Signals (HSAD)*. arXiv:2509.13154, 2025.